

Part 9: The Data Warehouse and What it Reports

PART 9: THE WAREHOUSE AND WHAT IT REPORTS

Problem Inventory

#	Problem	Financial Impact	Root Cause
9.1	Revenue excludes refunds	8.4% overstatement (~PKR 340M)	ETL doesn't join Refunds table
9.2	Revenue includes cancelled orders	1-3 day inflation	Cancellation job runs after ETL
9.3	Revenue excludes promotional discounts	Unknown % overstatement	Promotions table never integrated
9.4	Customer_Dim is Type 1 SCD — overwrites city	14% Karachi overstatement, 19% Peshawar understatement	No historical tracking
9.5	Tax filing uses order_date not payment_received_date	PKR 48M Q3 discrepancy	Wrong date dimension join
9.6	Basket size understated 34%	Store manager commissions underpaid ~PKR 12M	POS transaction fragmentation
9.7	Inventory turnover understated 40%	Over-ordering, 22% storage cost increase	ETL snapshots at 2AM daily max
9.8	NovaPay default rate: reported 7.8% vs actual 3.2%	PKR 420M over-provisioned reserves	No distinction: default vs restructured vs reversed

- **9.1 Revenue excludes refunds**

- **Problem:** The ETL pipeline fails to join the Refunds table, resulting in an 8.4% overstatement of revenue (~PKR 340M).
- **Solution:** Modify the ETL scripts to actively join the Refunds table and deduct the amounts during the net revenue calculation.
- **Why Chosen:** Directly aligns warehouse reporting with actual financial reality, satisfying critical accounting and compliance standards.

- **9.2 Revenue includes cancelled orders**

- **Problem:** The cancellation job runs after the ETL process, temporarily inflating daily revenue figures for 1-3 days.
- **Solution:** Reorder the ETL scheduling so cancellations are fully processed before the Sales_Fact table is populated.
- **Why Chosen:** Ensures the warehouse only captures finalized, immutable sales data, eliminating temporary reporting artifacts.
- **9.3 Revenue excludes promotional discounts**
 - **Problem:** Order promotions are never integrated into the warehouse, leading to an unknown overstatement of gross revenue.
 - **Solution:** Update the ETL to join the Order_Promotions and Promotions tables to properly deduct discount amounts.
 - **Why Chosen:** Captures the true net revenue by accounting for all marketing incentives, fixing distorted profit margins.
- **9.4 Customer_Dim is Type 1 SCD (overwrites city)**
 - **Problem:** Customer addresses are overwritten when they move, breaking historical geographic revenue tracking (e.g., misattributing Peshawar sales to Karachi).
 - **Solution:** Convert Customer_Dim to a Type 2 Slowly Changing Dimension (SCD) by adding effective_from and effective_to dates.
 - **Why Chosen:** Preserves the historical truth of where the customer lived at the exact time of the order, allowing accurate point-in-time regional analytics.
- **9.5 Tax filing uses order_date vs payment_received_date**
 - **Problem:** Tax reporting is legally misaligned because the warehouse relies purely on order_date instead of the legally required payment_received_date.
 - **Solution:** Add a Payment_Time_Dim to the warehouse and expose both date dimensions simultaneously in the Sales_Fact table.
 - **Why Chosen:** Supports dual-axis reporting, satisfying both operational metrics (when orders happen) and strict tax compliance (when money actually moves).

